Two dimensional Random Variable

1. Suppose that 2 cards are drawn at random from a pack of cards. Let X be the number of aces and Y be the number of queens obtained.

a) Find joint pdf of X and Y.

b) The marginal pdf of X and Y

c) Find $P(X \leq 1, Y > 1)$

d) Find $P(X+Y > 1)$

e) Conditional probability distribution of X given $Y=1$

f) Conditional probability distribution of Y given $X=1$

Sol^n :- Total 52 Cards.

Total no of aces $= 4$

Total no of queens $= 4$

From a standard deck two cards are drawn at random without replacement.

$\therefore$ Sample Size $= \binom{52}{2} = 1326$

a) $X : \{0,1,2\}$ No aces in 2 cards $Y : \{0,1,2\}$ No of queens in 2 cards.

$P(X=0, Y=0) = P(\text{No ace, No queen})$

$$= \frac{{}^{44}C_2}{{}^{52}C_2} = \frac{473}{663}$$

$P(X=0, Y=1) = P(\text{No ace, 1 queen})$

$$= \frac{{}^{4}C_1 \times {}^{44}C_1}{{}^{52}C_2} = \frac{88}{663}$$

$P(X=0, Y=2) = P(\text{No ace, 2 queens})$

$$= \frac{{}^{4}C_2 \times {}^{44}C_0}{{}^{52}C_2} = \frac{1}{221}$$

$P(X=1, Y=0) = P(\text{1 ace, 0 queens})$

$$= \frac{{}^{4}C_1 \times {}^{44}C_1}{{}^{52}C_2} = \frac{88}{663}$$

$P(X=2, Y=0) = P(\text{2 ace, 0 queens})$

$$= \frac{{}^{4}C_2 \times {}^{44}C_0}{{}^{52}C_2} = \frac{1}{221}$$

$P(X=1, Y=1) = P(\text{1 ace, 1 queen})$

$$= \frac{{}^{4}C_1 \times {}^{4}C_1 \times {}^{44}C_0}{{}^{52}C_2}$$

$$= \frac{8}{663}$$

And all other pairs have 0 probability.
Because $X+Y \leq 2$.

Joint probability distribution

X \ Y	0	1	2	P(X)
0	$\frac{473}{663}$	$\frac{88}{663}$	$\frac{1}{221}$	$\frac{188}{221}$
1	$\frac{88}{663}$	$\frac{8}{663}$	0	$\frac{32}{221}$
2	$\frac{1}{221}$	0	0	$\frac{1}{221}$
P(Y)	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$	

b) Marginal distribution of X

X :	0	1	2
P(X) :	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

Y :	0	1	2
P(Y) :	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

c) $P(X \leq 1, Y > 1) = P(X = 0, 1, Y = 2)$

$= P(0, 2) + P(1, 2)$

$= \frac{1}{221} + 0 = \frac{1}{221}$

d) $P(X+Y>1) = 1 - P(X+Y \le 1)$

$= 1 - \{P(0,0) + P(1,0) + P(0,1)\}$

$= 1 - \left\{ \frac{473}{663} + \frac{88}{663} + \frac{88}{663} \right\}$

$= 1 - \frac{649}{663} = \frac{14}{663}$

e) $P(X|Y=1) = P(X=0,1,2 \mid Y=1)$

$= \frac{P((X=0,1,2) \cap Y=1)}{P(Y=1)}$

X : 0 1 2

$P(X|Y=1)$: $\frac{\frac{88}{663}}{\frac{32}{221}}$ $\frac{\frac{8}{663}}{\frac{32}{221}}$ $\frac{0}{\frac{32}{221}}$

X :	0	1	2
$P(X\|Y=1)$:	$\frac{88}{96}$	$\frac{8}{96}$	0

f) $P(Y|X) = \frac{P(Y,X)}{P(X)} = \frac{P(X,Y)}{P(X)}$

$x \backslash y$	0	1	2
0	$\dfrac{\frac{473}{663}}{\frac{188}{221}} = \dfrac{473}{564}$	$\dfrac{\frac{88}{663}}{\frac{188}{221}} = \dfrac{22}{47}$	$\dfrac{\frac{1}{121}}{\frac{188}{221}} = \dfrac{1}{188}$
1	$\dfrac{\frac{88}{663}}{\frac{32}{221}} = \dfrac{11}{12}$	$\dfrac{\frac{8}{663}}{\frac{32}{221}} = \dfrac{1}{12}$	0
2	$\dfrac{\frac{1}{121}}{\frac{1}{121}} = 1$	0	0

Problems on expectation, variance, covariance and correlation.

P_{XY} =	$X \setminus Y$	0	1
	0	0.1	0.2
	1	0.3	0.4

Marginal distribution for X

$P(X=0) = P_{XY}(0,0) + P_{XY}(0,1) = 0.1+0.2=0.3$

$P(X=1) = P_{XY}(1,0) + P(_{XY})(1,1) = 0.3+0.4=0.7$

Marginal distribution for Y

$P(Y=0) = P_{XY}(0,0) + P_{XY}(1,0) = 0.1+0.3=0.4$

$P(Y=1) = P_{XY}(0,1) + P_{XY}(1,1) = 0.2+0.4=0.6$

<u>Expectation</u>

$E[X] = \sum_x x \, P(X=x)$ $\quad E[Y] = \sum_y y \, P(Y=y)$

$E[X] = 0 \; P(X=0) + 1 \; P(X=1)$

$= 0 \times 0.3 + 1 \times 0.7 = 0.7$

$E[Y] = 0 \; P(Y=0) + 1 \; P(Y=1)$

$= 0 \times 0.4 + 1 \times 0.6 = 0.6$

Variance

$$Var[X] = E[X^2] - (E[X])^2$$

$$E[X^2] = 0^2 P(X=0) + 1^2 P(X=1) = 0.7$$

$$Var[X] = 0.7 - (0.7)^2 = 0.21$$

$$Var[Y] = E[Y^2] - (E[Y])^2$$

$$E[Y^2] = 0^2 P(Y=0) + 1^2 P(Y=1)$$
$$= 0.6$$

$$Var[Y] = 0.6 - (0.6)^2 = 0.24$$

Standard deviation

$$\sigma_X = \sqrt{0.21} = 0.458$$
$$\sigma_Y = \sqrt{0.24} = 0.490$$

Covariance :-

$$E[XY] = \sum_x \sum_y xy \, P_{XY}(x, y)$$

X	Y	$P(x,y)$	$xy\ P(X,Y)$
0	0	0.1	0
0	1	0.2	0
1	0	0.3	0
1	1	0.4	0.4

$$E[XY] = 0+0+0+0.4 = 0.4$$

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

$$= 0.4 - (0.7)(0.6)$$

$$= -0.02$$

$\Rightarrow$ X and Y tend to vary in opposite directions

Correlation

$$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)\ Var(Y)}}$$

$$= \frac{-0.02}{\sqrt{0.21 \times 0.24}} = -0.0891$$

$\Rightarrow$ X and Y have a very weak inverse relationship.